

Rules, Workflow, Approval & Automation Engine

Functional Design & Backend Screens

TICVAI | Professional Reference Document

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Board 2 — Final Screen Register

Board 1 — Configure & Govern

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Next: Area 13 — Rules, Workflow, Approval & Automation Engine

This is an important platform-level area because we have repeatedly used **approval workflows, configurable business rules, exceptions, triggers, and AI-assisted automation** throughout TICVAI. These should not be independently hard-coded inside every module.

I recommend 2 boards / 20 backend screens:

Board Scope

Screens

Board 1 Business Rules, Workflow & Approval Configuration

10

Board 2 Workflow Operations, Automation, Monitoring & AI

10

Board 1 — Configure the Logic

The screenshot displays the TICVAI Rules & Workflow Configuration interface, organized into several functional modules:

- 13.1.1 Rules & Workflow Command Center:** Provides an overview of active rules, workflows, and pending approvals. It includes filters for status (Active, Draft, Pending) and counts for each.
- 13.1.2 Visual Business Rule Builder:** A tool for creating and editing business rules. It features a rule editor with conditions and actions, and a visual rule builder for complex logic.
- 13.1.3 Conditions, Decision Logic & Decision Tables:** A module for defining decision logic using decision tables and logic builders. It includes a decision table editor and a logic builder.
- 13.1.4 Visual Workflow Designer:** A visual tool for designing workflows. It includes a workflow editor with a visual canvas for mapping out the process flow.
- 13.1.5 Approval Matrix & Multi-Level Approval Configuration:** A module for configuring approval matrices and multi-level approval processes. It includes an approval matrix editor and a multi-level approval configuration tool.
- 13.1.6 Roles, Authority, Delegation & Approval Limits:** A module for defining roles, authority, delegation, and approval limits. It includes a role editor and a delegation configuration tool.
- 13.1.7 SLA, Escalation, Reminder & Timeout Rules:** A module for configuring SLA, escalation, reminder, and timeout rules. It includes an SLA editor and a reminder configuration tool.
- 13.1.8 Trigger, Action & Cross-Module Orchestration Configuration:** A module for configuring triggers, actions, and cross-module orchestration. It includes a trigger editor and an action configuration tool.
- 13.1.9 Workflow Testing, Simulation & Impact Analysis:** A module for testing, simulating, and analyzing the impact of workflows. It includes a workflow tester and a simulation tool.
- 13.1.10 Versioning, Governance, Approval & Publication:** A module for managing workflow versions, governance, approval, and publication. It includes a version manager and a governance tool.

The 10 screens would cover:

Rules & Workflow Command Center → Business Rule Builder → Conditions & Decision Logic → Workflow Designer → Approval Matrix → Roles & Authority Rules → Escalation & SLA Rules → Trigger & Action Configuration → Testing & Simulation → Versioning, Approval & Publication

Example:

Refund > AED 1,000
 AND Event Status ≠ Cancelled
 → Supervisor Approval
 → Finance Approval
 → Execute Refund

Or:

Discount > 15%
 → Commercial Director approval.

Or:

Purchase Order > AED 100,000
→ CFO approval.

Board 2 — Execute & Automate

This would cover:

Workflow Operations Center → Approval Inbox → Workflow Instance Monitor → Exceptions & Failed Workflows → Escalation Monitor → Automation Execution → Cross-Module Workflow Orchestration → Workflow Analytics → Process Optimization → AI Automation Intelligence

This is where TICVAI can become particularly powerful.

For example, AI could detect:

73% of refund approvals below AED 250 have been approved without modification during the last six months.

Then recommend:

Consider changing this workflow from **Manager Approval** to **Auto-Approve under AED 250**, subject to configured conditions.

Why centralize this

Without a central engine, we risk building:

**Ticketing Approval Engine
Finance Approval Engine
CRM Approval Engine
Resource Approval Engine
Procurement Approval Engine
Customer Service Approval Engine**

That becomes extremely difficult to govern.

Instead:

Ticketing | Finance | CRM | F&B | Retail | Resource | Group Sales | Customer Service | Wallet

↓

TICVAI Rules, Workflow & Automation Engine

↓

Evaluate → Route → Approve → Escalate → Execute → Audit

Each module still owns its business data and actions, but the common workflow/orchestration capability is centralized.

So I recommend our next step is **Area 13 → Board 1 — Business Rules, Workflow & Approval Configuration — 10-screen SOW.**

Start with Board 1 scope of work.

Area 13 — Rules, Workflow, Approval & Automation Engine

Board 1 — Business Rules, Workflow & Approval Configuration

10 Backend Screens

Board Objective

Board 1 creates the centralized **no-code/low-code business rules and workflow configuration engine** for TICVAI.

Instead of every TICVAI module building its own approval and workflow logic, modules should consume a common platform service.

The configuration journey is:

Discover → Define Rule → Build Logic → Design Workflow → Configure Approval → Define Authority → Configure SLA/Escalation → Connect Actions → Test → Govern & Publish

The engine should support workflows across:

Ticketing | Pricing | Finance | Procurement | CRM | F&B | Retail | Resource Management | Group Sales | Customer Service | Membership | Wallet | Waiver | Subscription & Licensing

Screen 13.1.1 — Rules & Workflow Command Center

Purpose

Provide administrators with a centralized portfolio of all business rules, workflows, approvals and automations configured across TICVAI.

Executive KPIs

Display:

- Active Rules
- Active Workflows
- Approval Workflows
- Draft Configurations
- Pending Approval
- Scheduled Changes
- Rules With Errors
- Workflows With Warnings
- Recently Modified
- Modules Covered

Portfolio Directory

Each configuration should show:

- Rule/Workflow ID
- Name
- Type
- Source Module
- Business Process
- Version
- Owner
- Effective Date
- Status
- Last Modified

- Usage

Types

Classify as:

- Business Rule
- Approval Workflow
- Operational Workflow
- Decision Rule
- Validation Rule
- Escalation Rule
- Automation
- Cross-Module Workflow

Example Records

REF-APP-001 — High Value Refund Approval

Customer Service → Refund

DISC-APP-004 — Discount Approval

Pricing → Commercial Approval

PO-APP-002 — Purchase Order Approval

Procurement → Finance

Status

Draft → Testing → Review → Pending Approval → Approved → Scheduled → Active → Suspended → Retired

Dependency Indicator

Example:

Used by 6 modules / 18 processes / 42 active configurations.

AI Capability

AI can detect duplicate or overlapping logic.

Three active refund workflows contain substantially similar approval conditions. Consider consolidating them into a reusable policy.

Acceptance Condition

Administrators can locate, understand and govern all TICVAI rules and workflows from one centralized workspace.

Screen 13.1.2 — Visual Business Rule Builder

Purpose

Allow administrators to create business rules without software development.

Rule Structure

Use:

WHEN / IF

→ Conditions

THEN

→ Outcome/Action

Example**IF**

Refund Amount > AED 1,000

AND

Event Status ≠ Cancelled

THEN

Require Supervisor Approval.

Visual Builder

I recommend a visual structure:

Trigger

↓

Conditions

↓

Decision

↓

Outcome**Condition Types****Support:**

- Equals
- Not Equals
- Greater Than
- Less Than
- Between
- Contains
- In List
- Exists
- Does Not Exist
- Before/After
- Percentage Threshold
- Boolean

Business Objects

Rules should consume governed fields from registered TICVAI modules.

Examples:

Order.TotalAmount

Refund.Amount

Customer.MembershipTier

Ticket.EventDate

Partner.CreditUtilization

PurchaseOrder.Value

Rule Outcomes**Examples:**

- Allow
- Reject
- Require Approval
- Require Additional Information
- Apply Hold
- Create Task
- Generate Alert
- Start Workflow
- Execute Approved Action

Rule Explanation

TICVAI should automatically provide a human-readable interpretation:

Refunds above AED 1,000 require supervisor approval unless the associated event has been cancelled.

Acceptance Condition

Authorized administrators can configure deterministic business decisions using governed TICVAI data without changing application code.

Screen 13.1.3 — Conditions, Decision Logic & Decision Tables

Purpose

Configure advanced decision logic where simple IF/THEN rules are insufficient.

Logical Operators

Support:

AND

OR

NOT

Nested Conditions

Example

Approve automatically if:

Refund $\leq$ AED 250

AND

Customer Verified = Yes

AND

(
Event Cancelled = Yes

OR

Service Failure Confirmed = Yes

)

Decision Tables

For complex scenarios, provide tabular configuration.

Example:

Refund	Customer Tier	Event Status	Result
≤ AED 250	Any	Cancelled	Auto Approve
AED 251-1K	Standard	Active	Supervisor
AED 251-1K	VIP	Active	Supervisor
> AED 1K	Any	Any	Manager + Finance

Rule Priority

When multiple rules apply:

- Priority
- Sequence
- Specificity
- Stop Processing
- Continue Evaluation

must be configurable.

Conflict Detection

TICVAI should detect:

- Contradictory Rules
- Overlapping Conditions
- Unreachable Outcomes
- Circular Logic
- Missing Outcomes

Reusable Rule Components

Examples:

High Value Transaction

VIP Customer

Event Within 24 Hours

Credit Limit Exceeded

These can be reused by multiple workflows.

AI Capability

AI can explain complex decision logic in plain language and identify potential conflicts.

Acceptance Condition

Complex business decisions can be modeled predictably, tested and reused without creating contradictory or ambiguous outcomes.

Screen 13.1.4 — Visual Workflow Designer

Purpose

Allow administrators to visually design complete business processes.

Workflow Canvas

Provide drag-and-drop nodes such as:

- Start
- Trigger
- Task
- Decision
- Approval
- System Action
- Notification
- Wait
- Timer
- Parallel Branch
- Merge
- Escalation
- Sub-Workflow
- End

Example

Refund Requested

↓

Validate Policy

↓

Refund > AED 1,000?

↙ YES ↘ NO

Supervisor Approval Auto Process

↓

Finance Approval

↓

Execute Refund

↓

Notify Customer

↓

Close

Workflow Properties

Configure:

- Workflow Name
- Module
- Business Process

- Owner
- Trigger
- Version
- Priority
- Effective Dates

Parallel Processing

Example:

After event cancellation:

→ Finance prepares refund

AND

→ Communication sends customer notification

AND

→ Ticketing invalidates tickets.

Sub-Workflows

Reusable processes should be callable from other workflows.

Example:

Standard Finance Approval

Validation

The designer should identify:

- Dead Ends
- Missing Outcomes
- Circular Loops
- Missing Assignee
- Invalid Actions

Acceptance Condition

Administrators can visually design governed end-to-end business processes including decisions, approvals, tasks and system actions.

Screen 13.1.5 — Approval Matrix & Multi-Level Approval Configuration

Purpose

Configure when approvals are required and who must approve.

Approval Types

Support:

- Single Approval
- Sequential Approval
- Parallel Approval
- Any-One Approval
- All-Must-Approve
- Conditional Approval

- Multi-Level Approval

Example — Discount

0-10%

No Approval

10.01-15%

Sales Manager

15.01-20%

Commercial Director

>20%

Commercial Director + CFO

Example — Procurement

PO ≤ AED 10K

Department Head

AED 10K-50K

Department Head → Finance

AED 50K-100K

Department Head → Finance → CFO

> AED 100K

Department Head → Finance → CFO → CEO

Approval Conditions

Can depend on:

- Amount
- Percentage
- Module
- Product
- Venue
- Department
- Customer Type
- Risk
- Exception Type
- Legal Entity

Approval Behavior

Configure:

- Sequential/Parallel
- Minimum Approvals
- Rejection Behavior
- Request Changes
- Delegate
- Reassign
- Skip Conditions

Approval Context

Approvers should receive the relevant business context rather than only:

“Approve AED 18,000?”

For example:

- Request
- Customer
- Transaction
- Original Value
- Proposed Value
- Margin Impact
- Reason
- Supporting Evidence

Acceptance Condition

TICVAI can dynamically determine the required approval path based on transaction context and configured authority rules.

Screen 13.1.6 — Roles, Authority, Delegation & Approval Limits

Purpose

Define **who has authority to perform or approve specific actions**.

This should work with TICVAI RBAC/PBAC rather than replace it.

Authority Configuration

Define by:

- User
- Role
- Position
- Department
- Venue
- Business Unit
- Legal Entity
- Region

Approval Limits

Example:

Sales Manager

Discount: ≤15%

Refund: ≤AED 1,000

Goodwill: ≤AED 500

Commercial Director

Discount: ≤25%

Refund: ≤AED 10,000

Delegation

Support:

CFO delegates approvals to Finance Director from 1–15 September.

Capture:

- Delegator
- Delegate
- Scope
- Start
- End
- Reason

Acting Role**Support temporary authority for:**

- Leave
- Travel
- Vacancy
- Event operations

Segregation of Duties**Critical workflows should support rules such as:**

Requester cannot approve own request.

or:

Refund creator and final approver must be different users.

Authority Validation**Before assigning an approval, verify:**

- User active
- Required role
- Authority amount
- Business scope
- Delegation validity

AI Capability

AI can identify authority gaps.

No active approver currently has authority for refunds above AED 50K for Venue B.

Acceptance Condition

Every approval is routed only to an authorized approver with valid authority for that transaction and organizational scope.

Screen 13.1.7 — SLA, Escalation, Reminder & Timeout Rules**Purpose**

Define how workflows behave when people or systems do not act within the expected time.

SLA Types**Configure:**

- Response SLA

- Approval SLA
- Task SLA
- Resolution SLA
- System Action Timeout

Example

Refund Approval

Target: 4 hours

At 50%:

→ Reminder Approver

At 75%:

→ Second Reminder

At 100%:

→ Escalate Manager

After 6 hours:

→ Escalate Finance Director

Time Calculation

Support:

- Calendar Hours
- Business Hours
- Working Days
- Venue Calendar
- Holiday Calendar

Escalation Actions

- Notify
- Reassign
- Escalate
- Add Approver
- Create Task
- Raise Priority
- Trigger Backup Workflow

Timeout Behavior

Examples:

Do Nothing

Escalate

Auto-Reject

Auto-Approve — only where explicitly permitted

Send to Exception Queue

Reminder Channels

Use Area 12 Communication Service for:

- Email
- Push

- In-App
- SMS where appropriate

AI Capability

AI can identify workflows where SLA settings are unrealistic based on actual historical processing times.

Acceptance Condition

Every time-sensitive workflow can automatically remind, escalate or safely handle overdue activities according to configurable SLA policies.

Screen 13.1.8 — Trigger, Action & Cross-Module Orchestration Configuration

Purpose

Define what starts a workflow and what TICVAI services may be called during execution.

Trigger Types

Support:

Business Event

RefundRequested

DiscountExceptionCreated

PurchaseOrderSubmitted

PartnerCreditExceeded

Data Condition

Balance > Limit

Schedule

Daily / Weekly / Effective Date

Manual

Authorized user starts workflow.

Action Registry

Actions can include:

- Create Approval
- Create Task
- Update Status
- Apply Hold
- Release Hold
- Create Notification
- Generate Document
- Execute Refund
- Update Allocation
- Activate Membership
- Suspend Partner
- Call Approved API/Service
- Start Sub-Workflow

Cross-Module Example**Group Booking Confirmed**

↓

Resource Management

→ Reserve guides

↓

Finance

→ Create deposit requirement

↓

Ticketing

→ Reserve ticket inventory

↓

Area 12

→ Send confirmation

Action Permissions

A workflow must not automatically gain permission simply because an action exists.

Each action must validate:

- Workflow Authority
- User/System Authority
- Module Policy
- Business Rule

Failure Handling**Configure:**

- Retry
- Rollback where supported
- Compensation Action
- Exception Queue
- Human Intervention

Idempotency**Repeated events should not accidentally create duplicate:**

- Refunds
- Orders
- Payments
- Approvals
- Tickets

Acceptance Condition

Workflows can securely orchestrate approved actions across TICVAI modules while preserving module ownership, authorization and transactional integrity.

Screen 13.1.9 — Workflow Testing, Simulation & Impact Analysis

Purpose

Allow administrators to test rules and workflows before they affect live operations.

This is a critical screen.

Test Modes

Support:

- Manual Test Case
- Sample Transaction
- Historical Replay
- Scenario Simulation
- Batch Test

Example Scenario

Input:

Refund: AED 1,500

Customer: Gold

Event: Active

Reason: Customer Request

Expected path:

Policy Validation

→ Supervisor Approval

→ Finance Approval

→ Refund

Simulation Output

Show:

- Rules Evaluated
- Conditions Matched
- Decisions
- Approval Path
- **Actions**
- **Notifications**
- SLA
- **Expected Outcome**

Historical Replay

Test a proposed workflow against historical transactions without executing actions.

Example:

Apply proposed refund workflow to the previous 90 days.

Result:

- 18,420 transactions evaluated
- 13,200 Auto Approved
- 4,680 Supervisor Approval

- 540 Finance Approval
- 0 execution changes made

Impact Analysis

Before publication, show:

This workflow is used by:

3 modules

8 venues

14 business processes

Estimated 12,400 monthly executions

Regression Testing

New versions should be compared with existing versions.

AI Capability

AI can generate edge-case scenarios automatically.

Examples:

- Missing approver
- Exact threshold value
- Delegated approver
- Event cancelled
- Duplicate trigger
- Service timeout

Acceptance Condition

No workflow needs to be tested for the first time in production; administrators can simulate logic, routing and expected impact safely before publication.

Screen 13.1.10 — Versioning, Governance, Approval & Publication

Purpose

Control how rules and workflows move safely from configuration into production.

Versioning

Example:

Refund Approval

v1.0 — Retired

v1.1 — Active

v2.0 — Draft

Change Record

Capture:

- Version
- Changed By
- Change Date
- Change Reason
- Business Owner

- Technical Owner
- Risk Classification
- Test Results
- Approval

Compare Versions

Show:

v1.1 ↔ v2.0

Highlight changes to:

- Conditions
- Thresholds
- Approvers
- **Actions**
- SLA
- Escalation
- Integrations

Governance Workflow

Recommended:

Draft

→ **Tested**

→ **Business Review**

→ **Technical Validation**

→ **Approval**

→ **Scheduled**

→ **Active**

Publication

Support:

- Publish Now
- Schedule
- Selected Tenant
- Selected Venue
- Selected Brand
- Controlled Rollout

Effective Dating

Example:

New 2027 approval matrix becomes active 1 January 2027 at 00:00.

Rollback

Authorized administrators should be able to restore an approved previous version where technically safe.

Kill Switch

Critical workflows should support immediate suspension.

Audit Trail

Record:

- Who created
- Who changed
- Who tested
- Who approved
- Who published
- When
- What changed

AI Governance Review

AI can highlight risk before publication:

Proposed change removes Finance approval for refunds between AED 5,000–10,000. Approximately 284 historical transactions would have followed a different approval path.

Acceptance Condition

Only tested, approved and version-controlled rules/workflows can become active, with complete auditability and controlled rollback/suspension capability.

Board 1 — Final Screen Register

Screen	Backend Screen	Primary Responsibility
13.1.1	Rules & Workflow Command Center	Central workflow portfolio
13.1.2	Visual Business Rule Builder	No-code rule creation
13.1.3	Conditions, Decision Logic & Decision Tables	Advanced decision logic
13.1.4	Visual Workflow Designer	End-to-end process design
13.1.5	Approval Matrix & Multi-Level Approval Configuration	Approval routing
13.1.6	Roles, Authority, Delegation & Approval Limits	Approval authority
13.1.7	SLA, Escalation, Reminder & Timeout Rules	Time-based governance
13.1.8	Trigger, Action & Cross-Module Orchestration	Platform orchestration
13.1.9	Workflow Testing, Simulation & Impact Analysis	Safe testing

Screen	Backend Screen	Primary Responsibility
13.1.10	Versioning, Governance, Approval & Publication	Production governance

Board 1 Architecture

The board follows one clear configuration journey:

Rule → Logic → Workflow → Approval → Authority → SLA → Action → Test → Govern → Publish

Important architectural boundary

The **Workflow Engine** should **orchestrate**; it should not own the underlying **business transaction**.

For example:

Workflow Engine determines: **“Finance approval is required.”**

But **Finance** owns the financial transaction.

Workflow Engine determines: **“Refund may now execute.”**

But **Payments/Order Management** executes the refund.

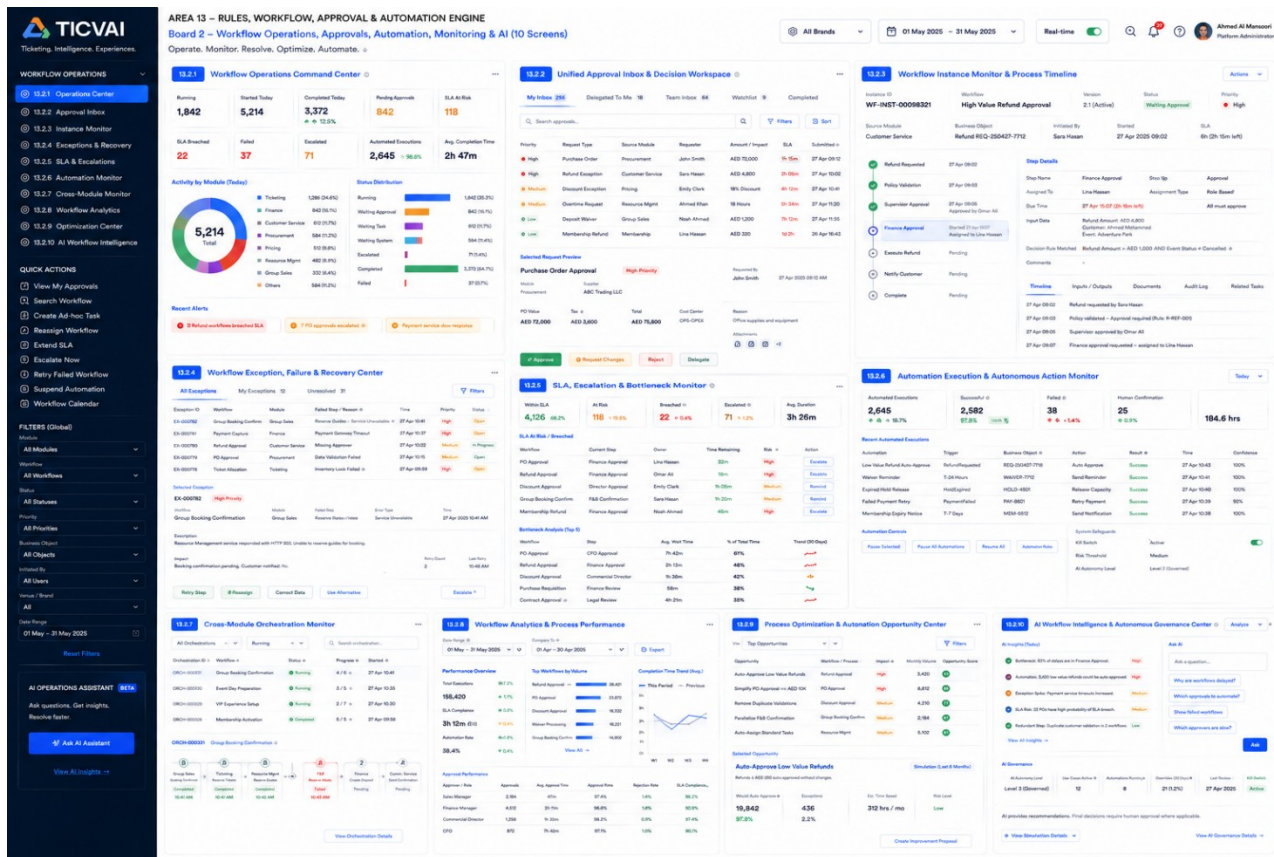
Workflow Engine determines: **“Send the customer confirmation.”**

But **Area 12 Communication Service** handles the delivery.

This separation is important because it gives TICVAI **one reusable orchestration brain across the platform** without creating a giant central module that takes ownership away from Ticketing, Finance, CRM, Resource Management and the other business domains.

Area 13 — Rules, Workflow, Approval & Automation Engine

Board 2 — Workflow Operations, Approvals, Automation, Monitoring & AI



1

0 Backend Screens

Board Objective

Board 1 configured the rules and workflows.

Board 2 is the **live operational layer** where TICVAI executes, monitors, manages, troubleshoots, analyzes, and optimizes those workflows across the entire platform.

The lifecycle is:

Monitor → Approve → Track → Resolve Exceptions → Escalate → Execute Automation → Orchestrate Modules → Analyze → Optimize → AI Govern

A critical distinction:

Board 1 = Design and configure how processes should work.

Board 2 = Operate and optimize processes while they are running.

Screen 13.2.1 — Workflow Operations Command Center

Purpose

Provide administrators and operational managers with a real-time view of all workflow activity across TICVAI.

Executive KPIs

Display:

- Workflows Running

- Started Today
- Completed Today
- Pending Approvals
- Waiting Tasks
- SLA At Risk
- SLA Breached
- Failed Workflows
- Escalated
- Automated Executions
- Average Completion Time
- Automation Success Rate

Module Overview

Show activity originating from:

- Ticketing
- Pricing
- Finance
- Procurement
- CRM
- Resource Management
- F&B
- Retail
- Group Sales
- Customer Service
- Membership
- Wallet
- Waiver
- Subscription & Licensing

Live Workflow Table

Display:

- Workflow Instance ID
- **Workflow**
- Module
- Business Object
- Initiated By
- Started
- Current Step
- Owner
- Priority
- SLA
- Status

Status

Use:

Running / Waiting Approval / Waiting Task / Waiting System / Escalated / Failed / Completed / Cancelled

Operational Health**Example:****Refund Approval Workflow**

842 running
 74 awaiting approval
 12 SLA risk
 3 failed

AI Capability**AI should summarize operational problems:**

63% of today's delayed workflows are waiting at Finance approval. The median approval time increased from 47 minutes to 2.3 hours.

Acceptance Condition

Authorized users can understand the real-time state and health of workflow execution across the entire TICVAI platform.

Screen 13.2.2 — Unified Approval Inbox & Decision Workspace

Purpose

Provide users with **one approval inbox across all TICVAI modules**.

This is extremely important.

A manager should not need to open Finance for one approval, Pricing for another, Procurement for another, and Customer Service for another.

Approval Inbox**Display:**

- Approval ID
- Request Type
- Source Module
- Requester
- Amount/Impact
- Priority
- Submitted
- SLA Remaining
- Approval Level
- Status

Examples**Pricing**

Discount Exception — 18%

Customer Service

Refund Exception — AED 4,800

Procurement

Purchase Order — AED 72,000

Resource Management

Overtime Request — 18 Hours

Approval Workspace

When opened, show:

Request

What is being requested?

Context

Why is approval required?

Financial/Operational Impact

What changes if approved?

Supporting Evidence

Documents, comments, history.

Approval Path

Where is this request in the workflow?

Actions**Support:****Approve****Reject****Request Changes****Delegate****Add Comment****View Full Record****Bulk Approval**

Allow only for approved low-risk categories.

Example:

Approve 18 routine purchase requests below AED 1,000.

AI Approval Summary**AI can summarize:**

Requested discount is 18%, 3% above the Sales Manager's authority. Expected contribution margin remains 31%. Customer generated AED 620K during the last 12 months.

Important AI Rule**AI may provide:**

- Context
- Comparison
- Risk
- Recommendation

But it must not hide relevant information from the approver.

Acceptance Condition

Authorized approvers can review and action approval requests from all TICVAI modules through one governed inbox.

Screen 13.2.3 — Workflow Instance Monitor & Process Timeline

Purpose

Allow administrators to inspect exactly what is happening inside an individual running workflow.

Instance Header

Display:

- Workflow Instance
- Workflow Name
- Version
- Source Module
- Business Object
- Initiated By
- Start Time
- Current Status
- Current Step
- SLA

Visual Process Map

Example:

Refund Requested ✓

↓

Policy Validation ✓

↓

Supervisor Approval ✓

↓

Finance Approval ← **CURRENT**

↓

Execute Refund

↓

Notify Customer

↓

Complete

Step Information

For each step show:

- Step
- Type
- Started
- Completed

- Assigned To
- Input
- Output
- Decision
- Duration
- Status

Decision Explanation

Example:

Finance approval was triggered because Refund Amount = AED 4,800 and configured threshold = AED 1,000.

Timeline

Show all:

- Rule evaluations
- Assignments
- Approvals
- Rejections
- Escalations
- **Notifications**
- API calls
- System actions
- Errors

Administrative Actions

Subject to permissions:

- Reassign
- Retry Step
- Skip Step where explicitly allowed
- Cancel Workflow
- Resume
- Escalate
- Open Source Record

Acceptance Condition

Administrators can reconstruct exactly how a workflow reached its current state and why every major decision or action occurred.

Screen 13.2.4 — Workflow Exception, Failure & Recovery Center

Purpose

Provide one controlled workspace for failed workflow executions.

Exception Categories

Support:

- Business Rule Failure
- Missing Data
- Missing Approver
- Permission Failure
- Integration Failure
- Timeout
- Action Failure
- Invalid State
- Duplicate Event
- Service Unavailable
- Configuration Error

Exception Queue

Display:

- Exception ID
- **Workflow**
- Instance
- Module
- Failed Step
- Error Type
- Time
- Retry Count
- Business Impact
- Priority
- Owner

Example

Workflow: Group Booking Confirmation

Failed Action: Reserve Guides

Reason: Resource Management unavailable

Booking: GRP-10822

Impact: Confirmation workflow incomplete.

Recovery Actions

Allow:

- Retry
- Retry From Step
- Reassign
- Correct Data
- Use Approved Alternative
- Resume
- Cancel
- Escalate

Dead-Letter Handling

Unrecoverable workflow events should enter a controlled exception queue rather than disappear.

Compensation Actions

Where a workflow partially executed, TICVAI may need to reverse previous actions.

Example:

Ticket inventory reserved ✓
Payment authorization failed ✕

Possible compensation:

Release reserved ticket inventory.

AI Root-Cause Detection

AI can group failures:

148 workflow failures during the last 20 minutes originate from the same Payment Service timeout.

Acceptance Condition

Failed workflows can be identified, investigated, recovered or safely compensated without losing business context or audit history.

Screen 13.2.5 — SLA, Escalation & Bottleneck Monitor

Purpose

Monitor workflows approaching or exceeding configured time limits.

SLA KPIs

Display:

- Within SLA
- At Risk
- Breached
- Escalated
- Average Processing Time
- Average Approval Time
- Longest Waiting Step

SLA Queue

Display:

- **Workflow**
- Instance
- Current Step
- Owner
- Started
- Target
- Time Remaining
- Risk
- Escalation Level

Bottleneck Analysis

Example:**Purchase Order Approval**

Average workflow: 11.2 hours

Breakdown:

- Department Head — 42 min
- Finance — 2.1 hrs
- CFO — **7.8 hrs**
- Execution — 18 min

TICVAI highlights:**Primary Bottleneck: CFO Approval****Escalation Monitor****Show:**

- First Reminder
- Second Reminder
- Manager Escalation
- Executive Escalation
- Final Outcome

Supervisor Actions**Authorized users can:**

- Reassign
- Escalate
- Extend SLA
- Add Backup Approver
- Change Priority

with mandatory reason capture where required.

AI Prediction**AI should identify likely breaches before they happen:**

22 purchase approvals have a 78%+ probability of missing SLA based on current queue and historical approver response time.

Acceptance Condition

Management can identify and resolve workflow bottlenecks before they materially impact business operations.

Screen 13.2.6 — Automation Execution & Autonomous Action Monitor

Purpose

Provide visibility and governance over actions executed automatically by TICVAI.

This becomes especially important as TICVAI becomes more AI-driven.

Automation Dashboard**Display:**

- Automated Actions Today

- Successful
- Failed
- Human Confirmation Required
- Reversed
- Suspended
- Estimated Manual Actions Avoided
- Estimated Time Saved

Automation Types

Examples:

- Auto-Approve
- Auto-Reject
- Auto-Assign
- Auto-Routing
- Auto-Notification
- Auto-Release Capacity
- Auto-Retry
- Auto-Reconciliation
- Auto-Activation
- Auto-Suspension

Execution Record

Display:

- Automation
- Trigger
- Business Object
- Rule
- Action
- Result
- Confidence where AI-assisted
- Execution Time
- Status

Example

Waiver Reminder Automation

Trigger: T-24 hours

Participants evaluated: 842

Incomplete: 118

Messages triggered: 118

Successful: 116

Failed: 2

Automation Control

Authorized administrators can:

- Pause Automation
- Resume

- Disable
- Inspect Rule
- Inspect Execution
- Review Failures
- Trigger Kill Switch

Guardrail

Every autonomous action must be traceable back to:

Trigger → Rule → Workflow Version → Authority → Action → Result

Acceptance Condition

TICVAI provides complete visibility and control over automated business actions, including immediate suspension where required.

Screen 13.2.7 — Cross-Module Orchestration Monitor

Purpose

Monitor complex workflows involving multiple TICVAI services.

Example — Group Booking Confirmation

Group Sales

Booking confirmed ✓

↓

Ticketing

Reserve 420 tickets ✓

↓

Resource Management

Reserve 4 guides ✓

↓

F&B

Reserve 420 meals ✓

↓

Finance

Create deposit requirement ✓

↓

Communication Service

Send confirmation ✓

Orchestration View

Show each participating module as a node.

For every node:

- Service
- Action
- Status
- Started

- Completed
- Duration
- Input/Output
- Retry

Cross-Module Status

Use:

Not Started / Running / Waiting / Successful / Failed / Compensated

Dependency View

Show dependencies such as:

Ticket inventory must be reserved before customer confirmation can be sent.

Partial Failure

Example:

Ticketing ✓

Resource Management ✓

F&B ×

Finance not started

System should clearly show whether the workflow:

- Waits
- Retries
- Rolls back
- Continues partially
- Requires human intervention

Distributed Trace

Technical administrators should be able to trace the same transaction across services using a common correlation/workflow ID.

AI Capability

AI can identify recurring cross-service bottlenecks.

Acceptance Condition

Complex cross-module business processes can be monitored as one business workflow rather than requiring administrators to inspect individual microservices independently.

Screen 13.2.8 — Workflow Analytics & Process Performance

Purpose

Measure how effectively TICVAI's business workflows are performing.

Core KPIs

Analyze:

- Workflow Volume
- Completion Rate
- Failure Rate
- Average Completion Time
- Approval Time

- Automation Rate
- Escalation Rate
- Rejection Rate
- Rework Rate
- SLA Compliance

Analyze By

- **Workflow**
- Module
- Venue
- Brand
- Business Process
- Department
- Approver
- User
- Date
- Transaction Value

Process Funnel

Example:

10,000 Refund Requests

↓

9,840 Validated

↓

4,200 Auto-Approved

↓

5,640 Approval Required

↓

5,410 Approved

↓

230 Rejected

↓

9,610 Successfully Processed

Approval Analytics

Measure:

- Average Approval Time
- Approval Rate
- Rejection Rate
- Request Changes Rate
- Delegation Rate

Workflow Comparison

Compare:

Workflow v1.4 vs v1.5

Example:**Average completion:**

v1.4 — 4.8 hours

v1.5 — 2.9 hours

Automation Benefit**Estimate:**

- Manual Steps Removed
- Processing Time Saved
- Workload Reduced
- Cost Saving where measurable

Acceptance Condition

Management can quantitatively evaluate whether workflows and automation are improving operational efficiency.

Screen 13.2.9 — Process Optimization & Automation Opportunity Center

Purpose

Identify business processes that should be simplified, redesigned or automated.

This is where TICVAI moves beyond simply running workflows.

Opportunity Categories**Identify:**

- Repetitive Approval
- Unnecessary Approval
- High Manual Work
- Excessive Rework
- Long Waiting Time
- Duplicate Steps
- High Failure Rate
- Low-Risk Manual Action
- Process Bottleneck

Opportunity Record**Display:**

- Process
- Module
- Monthly Volume
- Current Steps
- Average Duration
- Manual Steps
- Approval Rate
- Exception Rate

- Estimated Opportunity

Example

Low-Value Refund Approval

Monthly Requests: 4,820

Value < AED 250: 3,420

Approval Rate: 99.2%

Average Approval Delay: 1.8 hours

Recommendation:

Consider auto-approval for qualifying low-risk refunds.

Simulation

Before changing anything:

Apply proposed automation to previous six months.

Estimated result:

- 19,840 approvals removed
- 97.8% would have followed same outcome
- 3,480 staff hours potentially saved
- 436 cases require exception handling

Human Review

Actions:

Simulate

Create Improvement Proposal

Send for Review

Create New Workflow Version

Acceptance Condition

TICVAI systematically identifies automation and simplification opportunities based on actual operational evidence rather than assumptions.

Screen 13.2.10 — AI Workflow Intelligence & Autonomous Governance Center

Purpose

Create the AI intelligence layer across TICVAI's entire rules, workflow and automation ecosystem.

This is the management-level AI brain for Area 13.

AI Inputs

Analyze:

- Rules
- Workflows
- Approvals
- Execution History
- SLA
- Exceptions

- Automation
- Process Performance
- Approver Behavior
- Module Activity
- Business Outcomes

Natural-Language Questions

Authorized users should be able to ask:

“Which workflows are causing the most delays?”

“Which approvals could potentially be automated?”

“Why are purchase orders taking longer this month?”

“Which workflows fail most often?”

“Show me approval bottlenecks above AED 50,000.”

“Which processes have unnecessary manual steps?”

AI Process Discovery

AI can analyze execution history and identify the actual process being followed.

Example:

Configured process:

Request → Manager → Finance → Execute

Actual common process:

Request → Manager → Request Changes → Manager → Finance → Finance Clarification → Execute

AI identifies excessive rework.

AI Recommendations

Automation

96.8% of refund requests below AED 200 are approved without modification.

Approval Simplification

Removing one approval layer for low-risk procurement could reduce median processing time by 3.4 hours.

SLA

Current Finance SLA of 2 hours is breached 41% of the time; historical median is 2.7 hours.

Workflow Redesign

Two sequential validation steps evaluate the same customer eligibility condition.

AI-Generated Workflow Proposal

AI may draft:

Suggested Refund Workflow v3.0

but it should **never publish it automatically**.

The proposal must enter:

Simulation → Business Review → Approval → Publication

AI Autonomy Levels

I recommend the same governance model across TICVAI:

Level 0 — Observe

AI only analyzes.

Level 1 — Recommend

AI proposes action.

Level 2 — Prepare

AI prepares workflow/action for human confirmation.

Level 3 — Governed Automation

Pre-approved deterministic/AI-assisted actions execute within defined limits.

Level 4 — Autonomous Low-Risk Operations

Only explicitly approved scenarios with monitoring and kill switch.

AI Governance Dashboard

For every AI-enabled workflow show:

- AI Model/Service
- **Use Case**
- Autonomy Level
- Decision Scope
- Confidence Threshold
- Human Approval Requirement
- Execution Volume
- Override Rate
- Exception Rate
- Last Review
- Owner

Explainability

For AI-assisted actions TICVAI should retain:

Input Context → Recommendation → Confidence → Applicable Rule → Human/System Decision → Final Action

Kill Switch

Provide:

Suspend AI Recommendations

Suspend AI Automation

Suspend Specific Use Case

without necessarily stopping deterministic workflows.

Acceptance Condition

TICVAI can use AI to discover, recommend and optimize workflow automation while retaining explainability, human governance, version control and immediate operational control.

Board 2 — Final Screen Register

Screen	Backend Screen	Primary Responsibility
13.2.1	Workflow Operations Command Center	Live workflow operations
13.2.2	Unified Approval Inbox & Decision Workspace	Platform-wide approvals
13.2.3	Workflow Instance Monitor & Process Timeline	Individual workflow tracing
13.2.4	Workflow Exception, Failure & Recovery Center	Failure recovery
13.2.5	SLA, Escalation & Bottleneck Monitor	Delay and SLA management
13.2.6	Automation Execution & Autonomous Action Monitor	Automation governance
13.2.7	Cross-Module Orchestration Monitor	Multi-module execution
13.2.8	Workflow Analytics & Process Performance	Performance analytics
13.2.9	Process Optimization & Automation Opportunity Center	Process improvement
13.2.10	AI Workflow Intelligence & Autonomous Governance Center	AI optimization/governance

Area 13 — Complete Architecture

With Board 2, **Area 13** contains 2 boards / 20 backend screens.

Board 1 — Configure & Govern

Rule → Logic → Workflow → Approval → Authority → SLA → Action → Test → Version → Publish

Board 2 — Execute & Optimize

Monitor → Approve → Trace → Recover → Escalate → Automate → Orchestrate → Analyze → Optimize → AI Govern

One architecture point I strongly recommend

The **Unified Approval Inbox (13.2.2)** should eventually also appear in the **TICVAI Employee App**.

A manager receives a push notification:

Approval Required — AED 72,000 Purchase Order

They open it and see:

Request → Business Context → Supporting Documents → Approval History → AI Summary

Then:

Approve / Reject / Request Changes

The same approval is immediately reflected in the backend Workflow Engine.

This would give TICVAI **one approval architecture across web backend + employee mobile app**, rather than building separate approval mechanisms for Finance, Procurement, Pricing, Customer Service, Group Sales and other modules.